

## BUSINESS REQUIREMENTS DOCUMENT

# FASOL DOCTOR

*AI-Powered Crop Disease Diagnosis for Smallholder Farmers*

Document Version: 1.0

Date: September 2026

## Table of Contents

---

|                                                |    |
|------------------------------------------------|----|
| 1. Executive Summary.....                      | 3  |
| 2. Problem Statement & Market Need.....        | 3  |
| 3. Project Objectives.....                     | 4  |
| 4. Project Scope.....                          | 4  |
| 5. Stakeholders & Target Users.....            | 5  |
| 6. Proposed Solution Overview.....             | 5  |
| 7. Core Features (MVP — Phase 1).....          | 5  |
| 8. Future / Advanced Features (Phase 2–3)..... | 6  |
| 9. Functional Requirements.....                | 6  |
| 10. Non-Functional Requirements.....           | 7  |
| 11. High-Level Technical Architecture.....     | 7  |
| 12. Business Model & Revenue Streams.....      | 8  |
| 13. Market Opportunity & Scalability.....      | 8  |
| 14. Success Metrics / KPIs.....                | 9  |
| 15. Risks & Mitigation.....                    | 9  |
| 16. Implementation Roadmap.....                | 9  |
| 17. Conclusion.....                            | 10 |

## 1. Executive Summary

---

Fasol Doctor is an AI-powered, voice-and-camera-based mobile and SMS platform that allows smallholder farmers in Bangladesh to diagnose crop pests and diseases within seconds and receive treatment guidance in Bangla — without needing to wait for or travel to reach a human agricultural expert.

Crop loss from undiagnosed pests and disease is one of the largest, most preventable drains on rural household income in Bangladesh. Fasol Doctor closes the gap between the moment a farmer notices a problem and the moment they receive reliable, actionable guidance, using computer vision, natural language processing, and localized agronomic data.

Beyond its direct social impact, Fasol Doctor is designed from day one as a sustainable, revenue-generating product: it creates a proprietary, hyper-local agricultural dataset that is valuable to agri-input companies, insurers, researchers, and government bodies, and it is directly scalable to other South Asian and African markets facing the same structural gap in agricultural extension services.

This document defines the business requirements, scope, features, and success criteria for the MVP and subsequent phases of Fasol Doctor, intended to guide design, development, and go-to-market decisions.

## 2. Problem Statement & Market Need

---

### 2.1 The Core Problem

Smallholder farmers make up the vast majority of Bangladesh's agricultural workforce, yet the ratio of government agricultural extension officers to farmers remains extremely low, especially outside district towns. When a crop shows signs of disease, pest infestation, or nutrient deficiency, most farmers currently rely on:

- Guesswork based on inherited, often outdated farming knowledge.
- Local input/pesticide sellers, who have a direct financial incentive to over-prescribe chemical inputs.
- Delayed visits from extension officers, by which point the disease may have already spread across the field.

Globally, pests and disease are estimated to destroy a significant share of crop yield every year, and the impact falls hardest on smallholders who cannot absorb the financial shock of a bad season.

### 2.2 Why Now

- Smartphone and mobile internet penetration in rural Bangladesh has crossed a critical threshold, making camera + voice-based tools viable at scale.
- Advances in lightweight, on-device computer vision models make offline-first diagnosis realistic even in low-connectivity areas.
- Bangla-language NLP and speech models have matured enough to support natural, low-literacy-friendly interaction.
- BRAC's existing field infrastructure (community health workers, microfinance officers, agriculture programs) provides a ready distribution and trust channel that most agri-tech startups do not have access to.

### 2.3 Who Is Affected

The primary affected group is smallholder and marginal farmers, particularly those farming rice, vegetables, and cash crops on small plots, who have the least buffer against yield loss. Secondary stakeholders affected

by this gap include agricultural input retailers (who currently absorb informal diagnostic responsibility they aren't equipped for), microfinance institutions with agri-loan portfolios exposed to crop failure risk, and government bodies tasked with food security monitoring.

### 3. Project Objectives

| # | Objective                                                                         | How Success Is Measured                                      |
|---|-----------------------------------------------------------------------------------|--------------------------------------------------------------|
| 1 | Enable instant, accurate crop disease/pest diagnosis for smallholder farmers      | Diagnostic accuracy rate; time from photo to recommendation  |
| 2 | Reduce crop yield loss attributable to late or incorrect treatment                | % reduction in reported yield loss (pilot vs. control group) |
| 3 | Reach farmers regardless of literacy, smartphone access, or connectivity          | % of interactions completed via SMS/voice vs. smartphone app |
| 4 | Build a sustainable, revenue-generating product — not a pure grant-dependent tool | Revenue from B2B/data partnerships within 18 months          |
| 5 | Create a scalable model exportable to other regions and crops                     | Number of new districts/countries onboarded post-pilot       |

### 4. Project Scope

#### 4.1 In Scope (MVP and Phase 2)

- Photo-based crop disease and pest identification for the top rice and vegetable diseases in the pilot region.
- Bangla voice and text symptom description as an input alongside or instead of photos.
- Treatment recommendations using locally available, low-cost inputs, with organic/low-chemical options prioritized where effective.
- Offline-first mobile app with SMS/USSD fallback for feature phones.
- A lightweight backend dashboard for BRAC field staff to monitor regional disease trends.

#### 4.2 Out of Scope (Initial Release)

- Diagnosis for crops outside the pilot's initial crop list (to be expanded in later phases).
- Direct e-commerce / marketplace for purchasing pesticides or fertilizers (considered for Phase 3).
- Drone or satellite-based field-level imagery analysis (considered for a future enterprise tier).
- Full integration with government agricultural databases (dependent on partnership agreements, not a technical blocker).

### 5. Stakeholders & Target Users

| Stakeholder                | Role / Interest                                                                       |
|----------------------------|---------------------------------------------------------------------------------------|
| Smallholder Farmers        | Primary end users; need fast, trustworthy, low-cost diagnosis and treatment guidance. |
| BRAC Agriculture Programme | Implementation partner; provides field distribution, trust, and domain expertise.     |

| Stakeholder                               | Role / Interest                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------|
| Community Agriculture Workers             | Intermediaries who can operate the tool on behalf of low-tech-literacy farmers.                  |
| Agri-input Companies                      | Potential data/advertising partners; interested in aggregated, anonymized demand signals.        |
| Microfinance / Insurance Providers        | Potential B2B customers interested in crop-risk data for agri-loans and crop insurance products. |
| Government (DAE, Ministry of Agriculture) | Potential long-term partner for outbreak monitoring and national-scale rollout.                  |
| Development Partners / Donors             | Interested in measurable social impact and cost-per-farmer-reached metrics.                      |

## 6. Proposed Solution Overview

Fasol Doctor is delivered through three coordinated channels designed to meet farmers where they already are:

- Mobile App (Android-first): camera-based diagnosis, Bangla voice interaction, offline caching of common diagnoses, and a personal “crop history” log.
- SMS / USSD Channel: for feature-phone users — farmers describe symptoms via a short menu or free text and receive a text-based recommendation, with an option to escalate to a human agent.
- Field Worker Console: a simplified tablet-based interface for BRAC's community agriculture workers to run diagnoses on behalf of farmers during home/field visits, feeding data back into the central system.

At the core is an AI diagnostic engine combining a computer-vision model fine-tuned on regional crop-disease image data with a Bangla-language conversational layer that captures symptoms the camera can't see (smell, texture, spread pattern, recent weather).

## 7. Core Features (MVP — Phase 1)

The following features form the minimum viable product to be validated in the initial pilot. Feature prioritization can be revisited with BRAC's programme team before development begins.

| Feature                      | Description                                                                             | Priority  |
|------------------------------|-----------------------------------------------------------------------------------------|-----------|
| Photo-based diagnosis        | Farmer photographs affected crop; AI returns likely disease/pest with confidence score. | Must-have |
| Bangla voice symptom input   | Farmer describes the issue by voice in natural Bangla; NLP extracts key symptoms.       | Must-have |
| Treatment recommendations    | Step-by-step, locally-available treatment options, ranked by cost and effectiveness.    | Must-have |
| Offline mode                 | Common disease models cached on-device; works without active internet connection.       | Must-have |
| SMS/USSD fallback            | Full diagnostic flow accessible via basic text messages for feature phones.             | Must-have |
| Confidence & escalation flag | Low-confidence cases are flagged for human expert review via                            | Must-have |

| Feature                          | Description                                                               | Priority    |
|----------------------------------|---------------------------------------------------------------------------|-------------|
|                                  | field workers.                                                            |             |
| Regional disease trend dashboard | Aggregated, anonymized view for BRAC staff to spot local outbreaks early. | Should-have |
| Farmer crop history log          | Personal record of past diagnoses and treatments per plot.                | Should-have |

## 8. Future / Advanced Features (Phase 2–3)

| Feature                                  | Description                                                                           | Target Phase |
|------------------------------------------|---------------------------------------------------------------------------------------|--------------|
| Weather-integrated risk alerts           | Proactive push/SMS alerts when weather conditions favor a known disease outbreak.     | Phase 2      |
| Marketplace integration                  | In-app ordering of recommended, verified-quality treatment inputs.                    | Phase 2      |
| Yield & loss prediction                  | AI-estimated yield impact if the issue is treated vs. left untreated.                 | Phase 2      |
| Insurance & agri-loan data API           | Licensed, anonymized risk data feed for microfinance and crop insurance underwriting. | Phase 3      |
| Multi-country / multi-language expansion | Model and language layer adapted for other South/Southeast Asian markets.             | Phase 3      |
| Drone/satellite field-level analysis     | Enterprise-tier plot-level monitoring for large cooperatives or agribusinesses.       | Phase 3      |

## 9. Functional Requirements

| ID    | Requirement                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-01 | The system shall accept an image of a crop/plant as diagnostic input.                                                                             |
| FR-02 | The system shall reject or flag unusable images (blurry, poorly lit, wrong subject) and prompt the farmer to re-capture before running diagnosis. |
| FR-03 | The system shall accept Bangla voice or text input describing symptoms.                                                                           |
| FR-04 | The system shall return a ranked list of likely diagnoses with confidence scores.                                                                 |
| FR-05 | The system shall provide treatment recommendations tied to each diagnosis, prioritizing locally available inputs.                                 |
| FR-06 | The system shall function without an internet connection for the most common disease models.                                                      |
| FR-07 | The system shall provide an SMS/USSD-based flow for non-smartphone users.                                                                         |
| FR-08 | The system shall flag low-confidence diagnoses for human expert review.                                                                           |

| ID    | Requirement                                                                                                                                                                                         |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-09 | The system shall log anonymized diagnosis data for regional trend analysis.                                                                                                                         |
| FR-10 | The system shall support field worker login, role-based access, and an assisted-diagnosis mode.                                                                                                     |
| FR-11 | The system shall allow farmers to view their past diagnosis and treatment history.                                                                                                                  |
| FR-12 | The system shall allow farmers (or field workers on their behalf) to mark whether a recommendation resolved the issue, feeding this outcome back into model retraining and treatment-ranking logic. |
| FR-13 | The system shall allow an authorized admin to add, edit, or deactivate entries in the treatment/disease database without requiring a code deployment.                                               |
| FR-14 | The system shall send proactive SMS/push notifications to farmers for follow-up reminders and, where integrated, weather-linked disease-risk alerts.                                                |
| FR-15 | The system shall support onboarding and authentication for new farmers and field workers, including a simplified flow for low-literacy users (e.g., phone-number + voice-based verification).       |
| FR-16 | The system shall allow authorized partners/admins to export anonymized, aggregated diagnostic and trend data in a structured format (e.g., CSV/API) for reporting and licensing.                    |
| FR-17 | The system shall support addition of new crops, diseases, and languages/dialects through configuration rather than core re-engineering.                                                             |
| FR-18 | The system shall display a clear disclaimer and human-escalation path when a diagnosis confidence score falls below a defined threshold.                                                            |

## 10. Non-Functional Requirements

| Category     | Requirement                                                                                                                     |
|--------------|---------------------------------------------------------------------------------------------------------------------------------|
| Performance  | Diagnosis result shall be returned within 5 seconds on-device (offline) and within 3 seconds online.                            |
| Usability    | Interface shall be usable by low-literacy users via icons, voice, and Bangla-only text.                                         |
| Availability | SMS/USSD channel shall maintain 99% uptime, independent of smartphone app status.                                               |
| Scalability  | Backend architecture shall support scaling from a pilot of ~5,000 to 1M+ farmers without redesign.                              |
| Data Privacy | Farmer-identifiable data shall not be shared externally without explicit consent; only anonymized/aggregated data is monetized. |
| Localization | All models and content shall be adaptable to new regional dialects, crops, and languages with minimal re-engineering.           |
| Security     | All data in transit and at rest shall be encrypted; field worker accounts shall be role-based and access-controlled.            |

## 11. High-Level Technical Architecture

The platform is structured in four layers to keep the system modular, allowing individual components (e.g., the vision model, or a single language pack) to be upgraded independently as the product scales:

- Input Layer: Mobile camera, Bangla voice/text, SMS/USSD gateway, and field worker console.

- **AI Core:** Computer-vision disease classifier + Bangla NLP symptom extractor + treatment recommendation engine, with an on-device lightweight model for offline use and a larger cloud model for connected sessions.
- **Data Layer:** Farmer profiles, diagnosis history, regional disease trend store, and an anonymization pipeline that separates personally identifiable data from data made available for analytics/licensing.
- **Interface & Distribution Layer:** Android app, SMS/USSD gateway integration with local telecom providers, and a web dashboard for BRAC staff and future institutional partners.

This layered design is deliberate: it is what allows the same core AI engine to later power a field-worker tool, a government dashboard, or a white-labeled product for another NGO or market — without being rebuilt from scratch.

## 12. Business Model & Revenue Streams

Fasol Doctor is designed to graduate from a grant/pilot-funded programme into a self-sustaining, and eventually profitable, product. Proposed revenue streams, in order of expected maturity:

| Revenue Stream                      | Description                                                                                                       | Customer                          |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| NGO / Development Partnerships      | Licensing the platform to other NGOs and development programmes for their own regions.                            | NGOs, donors                      |
| Anonymized Agri-Data Licensing      | Aggregated, anonymized disease/pest trend data sold as a subscription feed.                                       | Agri-input companies, researchers |
| Crop-Risk Data API                  | Structured risk signals for underwriting agri-loans and crop micro-insurance.                                     | MFIs, insurers                    |
| Sponsored / Verified Input Listings | Paid, clearly labeled placement for verified-quality treatment products (never influencing the diagnosis itself). | Agri-input companies              |
| Government / Enterprise Tier        | Custom deployment, dashboards, and outbreak-alerting for national agriculture bodies or agribusinesses.           | Government, large cooperatives    |

Critically, the diagnostic recommendation itself always remains free and unbiased for farmers — monetization sits entirely on top of the data and B2B layers, which protects farmer trust while still building a viable commercial model.

## 13. Market Opportunity & Scalability

Bangladesh's agricultural sector is overwhelmingly composed of small and marginal landholdings, meaning the addressable user base for a low-cost, accessible diagnostic tool is extremely large. Comparable AI crop-diagnosis products in other markets (for example, Plantix, which operates across India, Africa, and other regions) have already demonstrated that this category can scale to tens of millions of users and attract global investment, validating both the problem and the model.

Because the core AI engine, data pipeline, and business model are not Bangladesh-specific, Fasol Doctor is structurally built for expansion:

- **Horizontal expansion:** new crops (jute, tea, potato, mango) beyond the initial pilot list.
- **Geographic expansion:** other South and Southeast Asian countries with similar smallholder-farming structures and BRAC's existing regional presence (e.g., parts of Africa, given BRAC's operations there).



- Vertical expansion: from pure diagnosis into adjacent markets — crop insurance data, agri-lending risk scoring, and input marketplaces.

This makes Fasol Doctor not just a hackathon project, but a credible foundation for a standalone agri-tech venture or a flagship, licensable product within BRAC's own innovation portfolio.

## 14. Success Metrics / KPIs

| Metric                    | Target (Pilot Phase)                                                  |
|---------------------------|-----------------------------------------------------------------------|
| Farmers reached           | 5,000+ farmers within first 6 months of pilot                         |
| Diagnostic accuracy       | ≥ 80% agreement with expert agronomist review on flagged sample cases |
| Time to recommendation    | Under 5 seconds on-device average                                     |
| SMS/USSD adoption         | ≥ 30% of total interactions, validating reach to feature-phone users  |
| Reported yield protection | Measurable reduction in self-reported crop loss vs. control group     |
| Cost per farmer reached   | Decreasing month-over-month as user base scales                       |
| B2B partnership pipeline  | At least 2 signed data/partnership discussions within 12 months       |

## 15. Risks & Mitigation

| Risk                                                    | Mitigation                                                                                                                |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Low diagnostic accuracy on rare or overlapping diseases | Route low-confidence cases to human expert review via field workers; continuously retrain model with verified field data. |
| Low smartphone penetration limiting reach               | SMS/USSD channel and field-worker-assisted mode ensure access without a smartphone.                                       |
| Farmer trust / adoption resistance                      | Leverage BRAC's existing trusted field presence for onboarding rather than a cold digital launch.                         |
| Data privacy concerns                                   | Strict anonymization pipeline; no farmer-identifiable data sold or shared externally without consent.                     |
| Seasonal / crop-specific data scarcity at launch        | Prioritize pilot crops and diseases with the most available regional image and case data first.                           |
| Over-reliance on chemical input recommendations         | Rank organic/low-chemical treatments alongside chemical options; keep recommendation logic independent of any sponsor.    |
| Connectivity gaps in remote areas                       | Offline-first on-device model for the most common diagnoses; SMS as a universal fallback.                                 |

## 16. Implementation Roadmap

| Phase                                 | Timeline   | Key Milestones                                                                    |
|---------------------------------------|------------|-----------------------------------------------------------------------------------|
| Phase 0 — Discovery & Data Collection | Month 1–2  | Partner with BRAC agronomists; collect/label regional crop-disease image dataset. |
| Phase 1 — MVP Build & Pilot           | Month 3–6  | Launch app + SMS channel in 1–2 districts with BRAC field worker support.         |
| Phase 2 — Validate & Expand Features  | Month 7–12 | Add weather alerts, expand crop coverage, onboard first B2B data partner.         |
| Phase 3 — Scale & Monetize            | Year 2     | National rollout, government/insurance partnerships, explore regional expansion.  |

## 17. Conclusion

Fasol Doctor directly addresses a large, measurable, and long-standing problem for Bangladesh's smallholder farmers using a technically achievable AI solution — not a speculative one. It is designed to work within the real constraints of rural connectivity and literacy, to build on BRAC's unique field distribution advantage, and to evolve from a pilot programme into a genuinely sustainable, revenue-generating platform with clear expansion paths.

We believe Fasol Doctor represents both a strong social-impact case and a credible commercial opportunity, and we welcome the chance to refine this BRD further with BRAC's programme and technical teams ahead of development.